

7-day and 28-day strength models (independent)

Each model built from its own 30 results with the same protocol · In scale · exhaustive AICc over 716 hierarchical models · generated 2026-09-29

7-day model · 10 parameters · 30 mixtures

$\ln(f_7) \sim \text{RCF} + \text{SS} + \text{Carbonation (4 levels)} + \text{RCF} \times \text{SS} + \text{SS}^2 + \text{Carbonated} \times \text{RCF} + \text{Carbonated} \times \text{SS}$

$\ln f = 2.7295 + 0.0602 \cdot A + 0.7545 \cdot D + 0.0484 \cdot C_{0.5h} - 0.0939 \cdot C_{1h} + 0.0570 \cdot C_{5h} - 0.1060 \cdot A \cdot D - 0.6777 \cdot D^2 + 0.1995 \cdot K \cdot A - 0.3358 \cdot K \cdot D$

$\ln(f_7) = 1.04787 + 0.00831352 \cdot \text{RCF} + 0.0605047 \cdot \text{SS} - 1.414 \times 10^{-4} \cdot \text{RCF} \cdot \text{SS} - 4.8191 \times 10^{-4} \cdot \text{SS}^2 + 0.0849998 \cdot C[0.5 \text{ h}] - 0.0572559 \cdot C[1 \text{ h}] + 0.0935896 \cdot C[5 \text{ h}] + 0.00997314 \cdot K \cdot \text{RCF} - 0.00895475 \cdot K \cdot \text{SS}$

R² / adj / pred
0.96
0.94 / 0.91

RMSE fitted / left out
1.5
2.1 MPa left out

Residual scatter
15 %
replicate batches 18 %

Lack of fit
0.82
vs replicate error (p)

RCF Hierarchy SS Robust Carbonation (4 levels) Hierarchy RCF × SS Moderate SS² Robust Carbonated × RCF Moderate Carbonated × SS Moderate

All selection criteria agree (AICc, BIC, backward elimination at α = 0.10 and 0.05). The carbonation terms are kept in 94 % of residual-bootstrap re-selections but in 33 % of subsamples of 24 mixtures, because they rest on the 8 uncarbonated mixtures.

28-day model · 9 parameters · 30 mixtures

$\ln(f_{28}) \sim \text{RCF} + A/B + \text{SS} + \text{Carbonation (4 levels)} + \text{RCF}^2 + (A/B)^2$

$\ln f = 3.0167 + 0.1507 \cdot A - 0.1560 \cdot C + 0.2329 \cdot D + 0.3169 \cdot C_{0.5h} - 0.2261 \cdot C_{1h} - 0.2102 \cdot C_{5h} - 0.2738 \cdot A^2 + 0.3172 \cdot C^2$

$\ln(f_{28}) = 75.6619 + 0.0485975 \cdot \text{RCF} + 0.00621023 \cdot \text{SS} - 322.446 \cdot A/B - 6.8438 \times 10^{-4} \cdot \text{RCF}^2 + 352.496 \cdot (A/B)^2 + 0.316901 \cdot C[0.5 \text{ h}] - 0.226146 \cdot C[1 \text{ h}] - 0.210182 \cdot C[5 \text{ h}]$

R² / adj / pred
0.71
0.60 / 0.43

RMSE fitted / left out
4.0
5.4 MPa left out

Residual scatter
23 %
replicate batches 19 %

Lack of fit
0.31
vs replicate error (p)

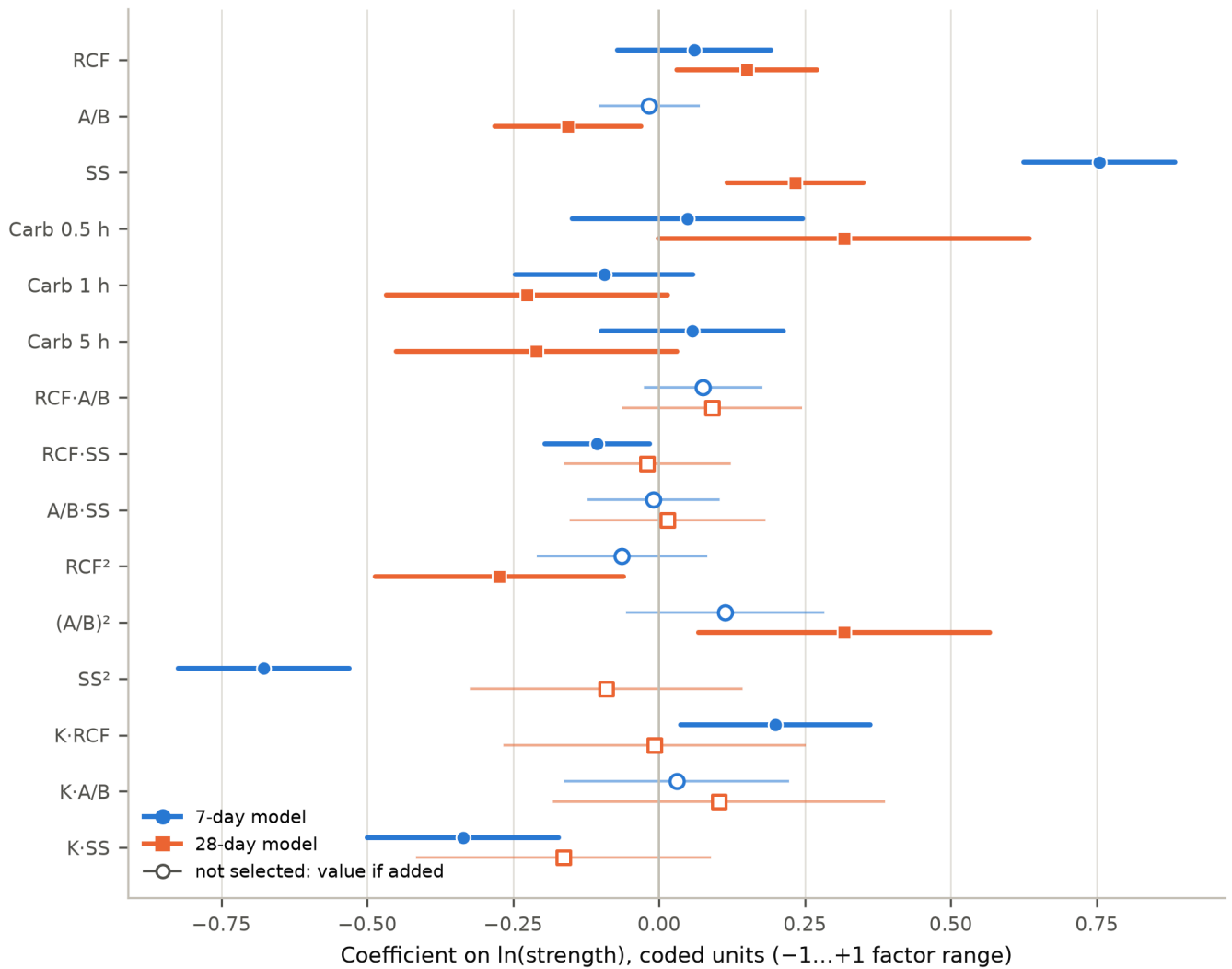
RCF Moderate A/B Moderate SS Robust Carbonation (4 levels) Moderate RCF² Moderate (A/B)² Moderate

SS is the only robust term. The selection is unstable: removing a single mixture changes the selected terms in 13 of 30 cases, subsamples of 24 mixtures re-select the model in 6 %, and nested cross-validation of the AICc procedure gives predicted R² −0.13 against 0.43 for the fixed terms. Carbonation enters as a level shift only (vs NC: 0.5 h ×1.37, 1 h ×0.80, 5 h ×0.81; carbonated levels differ, p = 0.005); the 0.5 h level rests on two compositions (mixes 1/12/19 and 10/14). Predictions for a new mixture are much less precise than at 7 days.

Terms side by side

Candidate term	7-day model: coefficient [95 % CI]	p	Evidence	28-day model: coefficient [95 % CI]	p	Evidence
RCF	0.060 [−0.071, 0.191]	0.35	Hierarchy	0.151 [0.031, 0.271]	0.016	Moderate
A/B	if added: −0.017 [−0.101, 0.067]	0.68	Not detected	−0.156 [−0.281, −0.031]	0.017	Moderate
SS	0.754 [0.625, 0.884]	< 0.001	Robust	0.233 [0.116, 0.350]	< 0.001	Robust
Carbonation (4 levels)	Carb 0.5 h: 0.048 [−0.149, 0.246] Carb 1 h: −0.094 [−0.246, 0.059] Carb 5 h: 0.057 [−0.099, 0.213]	0.22	Hierarchy	Carb 0.5 h: 0.317 [−0.001, 0.635] Carb 1 h: −0.226 [−0.467, 0.015] Carb 5 h: −0.210 [−0.450, 0.030]	0.008	Moderate
RCF × A/B	if added: 0.076 [−0.023, 0.175]	0.28	Not detected	if added: 0.091 [−0.061, 0.243]	0.23	Not detected
RCF × SS	−0.106 [−0.196, −0.016]	0.023	Moderate	if added: −0.020 [−0.161, 0.121]	0.77	Not detected
A/B × SS	if added: −0.010 [−0.120, 0.101]	0.90	Not detected	if added: 0.014 [−0.151, 0.180]	0.86	Not detected
RCF ²	if added: −0.064 [−0.208, 0.080]	0.37	Not detected	−0.274 [−0.486, −0.061]	0.014	Moderate
(A/B) ²	if added: 0.113 [−0.054, 0.281]	0.35	Not detected	0.317 [0.068, 0.566]	0.015	Moderate
SS ²	−0.678 [−0.824, −0.531]	< 0.001	Robust	if added: −0.090 [−0.322, 0.141]	0.43	Not detected
Carbonated × RCF	0.199 [0.037, 0.362]	0.019	Moderate	if added: −0.008 [−0.265, 0.249]	0.95	Not detected
Carbonated × A/B	if added: 0.030 [−0.161, 0.221]	0.87	Not detected	if added: 0.103 [−0.180, 0.385]	0.46	Not detected
Carbonated × SS	−0.336 [−0.500, −0.172]	< 0.001	Moderate	if added: −0.164 [−0.414, 0.087]	0.19	Not detected

Coefficients of the two independent models, 95 % CI

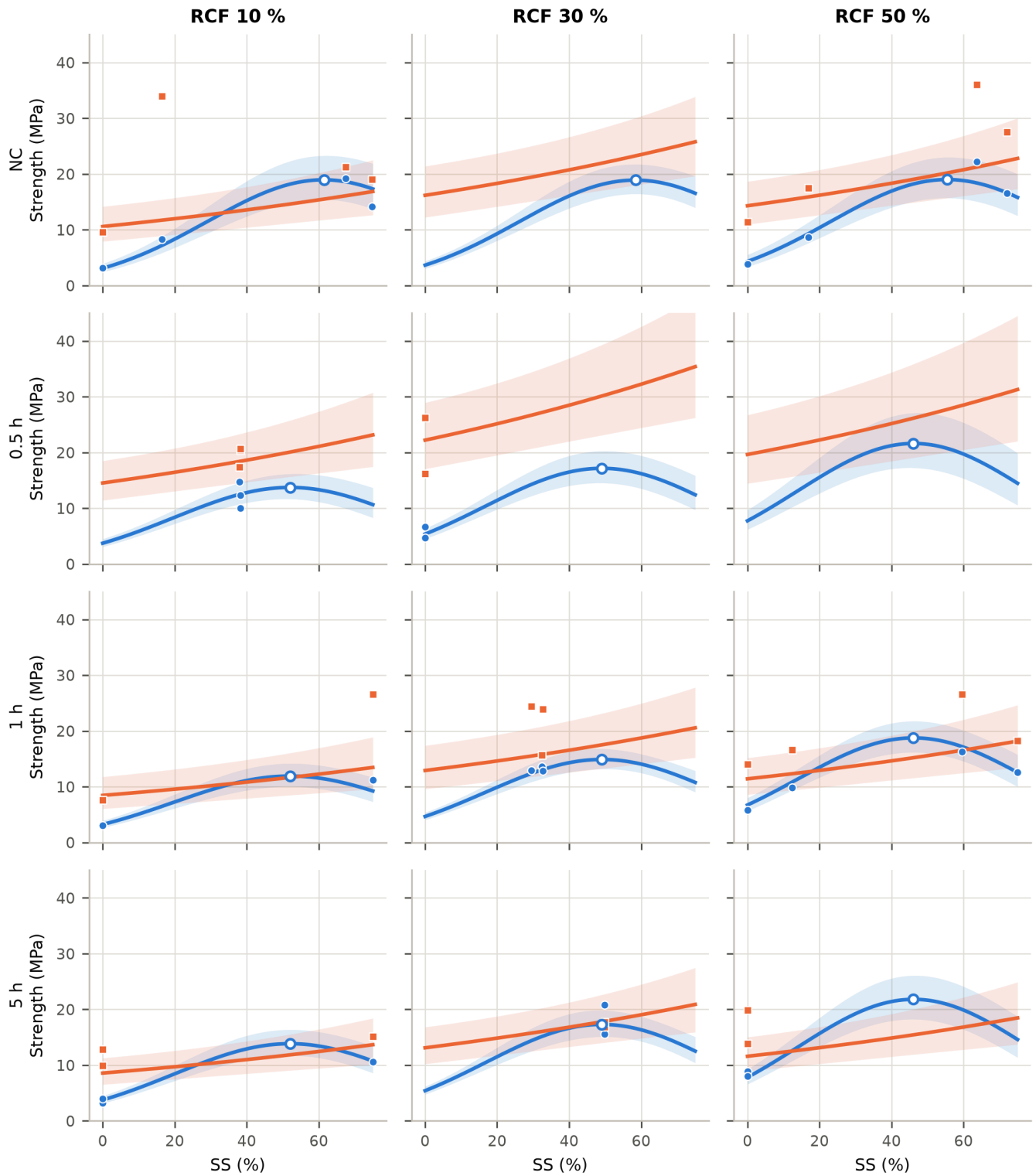


Trends side by side

SS

Carbonation	RCF (%)	7 d: SS at max [95 % boot]	28 d: SS at max	7 d: f(SS 0) / f(max)	28 d: f(SS 0) / f(max)	7 d: f(SS/SH = 2) / f(max)	28 d: f(SS/SH = 2) / f(max)	Suppl.: 28-day data, 7-day terms: SS at max
NC	10	61 % [52-72]	75 % (edge)*	×0.16 [0.12-0.21]	×0.63 [0.50-0.79]	×0.99 [0.95-1.02]	×0.95 [0.93-0.97]	65 %
NC	30	58 % [52-67]	75 % (edge)*	×0.19 [0.15-0.24]	×0.63 [0.50-0.79]	×0.97 [0.92-1.02]	×0.95 [0.93-0.97]	64 %
NC	50	56 % [46-64]	75 % (edge)*	×0.23 [0.18-0.30]	×0.63 [0.50-0.79]	×0.94 [0.87-1.01]	×0.95 [0.93-0.97]	62 %
1 h	10	52 % [42-58]	75 % (edge)*	×0.27 [0.22-0.33]	×0.63 [0.50-0.79]	×0.90 [0.82-0.99]	×0.95 [0.93-0.97]	52 %
1 h	30	49 % [42-54]	75 % (edge)*	×0.31 [0.27-0.36]	×0.63 [0.50-0.79]	×0.86 [0.78-0.95]	×0.95 [0.93-0.97]	50 %
1 h	50	46 % [40-51]	75 % (edge)*	×0.36 [0.30-0.43]	×0.63 [0.50-0.79]	×0.82 [0.72-0.92]	×0.95 [0.93-0.97]	49 %

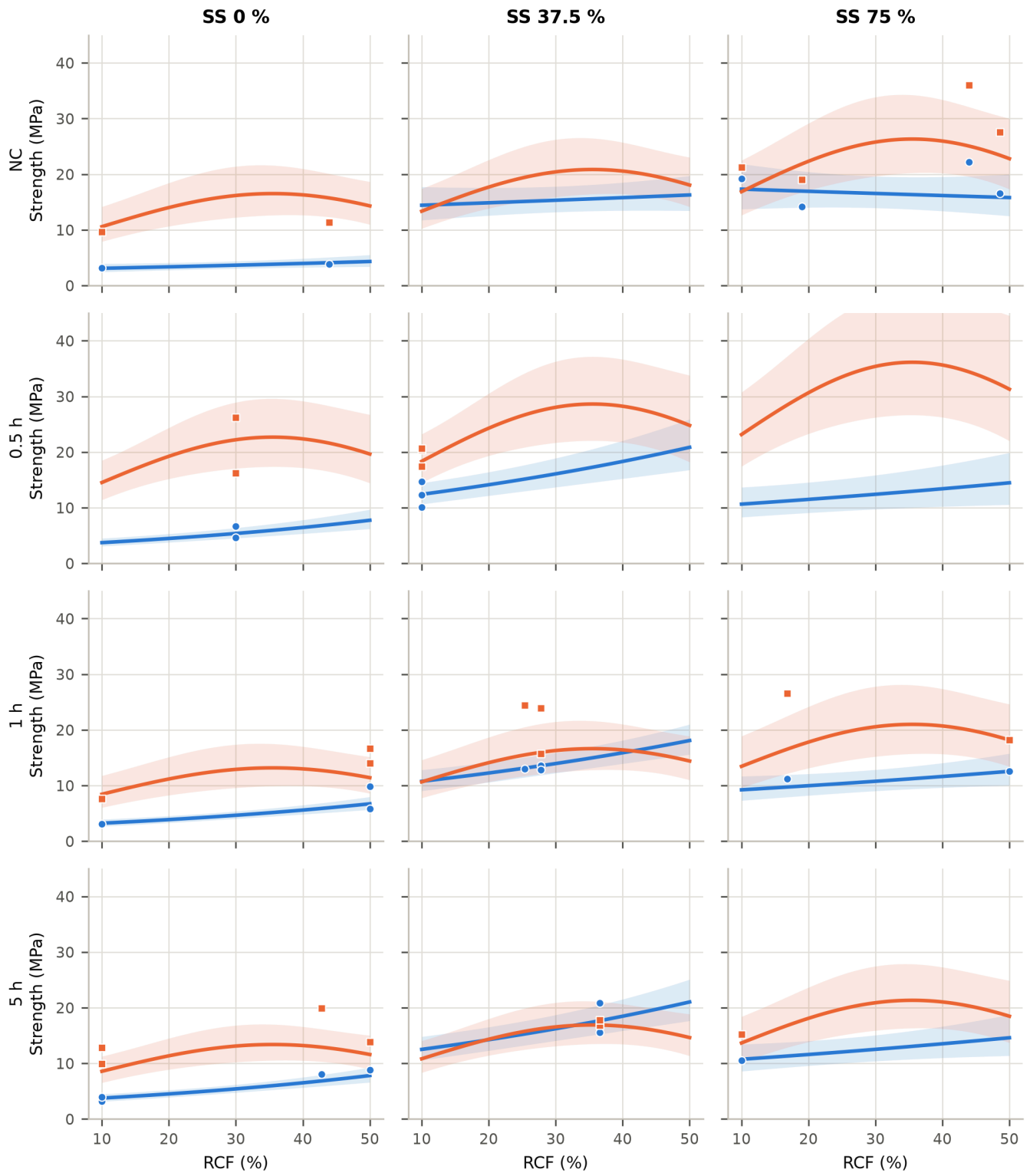
— 7-day model (95 % CI) ● runs within $\pm 10\%$ RCF
 — 28-day model (95 % CI) ○ interior peak



RCF 50 % / 10 %

Carbonation	SS (%)	7-day model	28-day model	Supplementary: 28-day data refitted with the 7-day terms
NC	0	$\times 1.39$ [1.01–1.92], $p = 0.041$	$\times 1.35$ [1.06–1.72], $p = 0.016$	$\times 1.23$ [0.64–2.36], $p = 0.52$
NC	25	$\times 1.21$ [0.93–1.58], $p = 0.15$	$\times 1.35$ [1.06–1.72], $p = 0.016$	$\times 1.20$ [0.69–2.09], $p = 0.51$
NC	50	$\times 1.05$ [0.80–1.38], $p = 0.70$	$\times 1.35$ [1.06–1.72], $p = 0.016$	$\times 1.17$ [0.67–2.04], $p = 0.57$
NC	75	$\times 0.91$ [0.66–1.25], $p = 0.55$	$\times 1.35$ [1.06–1.72], $p = 0.016$	$\times 1.14$ [0.59–2.20], $p = 0.68$
1 h	0	$\times 2.08$ [1.66–2.61], $p < 0.001$	$\times 1.35$ [1.06–1.72], $p = 0.016$	$\times 1.40$ [0.88–2.23], $p = 0.15$
1 h	25	$\times 1.80$ [1.50–2.17], $p < 0.001$	$\times 1.35$ [1.06–1.72], $p = 0.016$	$\times 1.36$ [0.93–2.00], $p = 0.11$
1 h	50	$\times 1.57$ [1.26–1.94], $p < 0.001$	$\times 1.35$ [1.06–1.72], $p = 0.016$	$\times 1.33$ [0.85–2.08], $p = 0.20$
1 h	75	$\times 1.36$ [1.01–1.83], $p = 0.042$	$\times 1.35$ [1.06–1.72], $p = 0.016$	$\times 1.30$ [0.70–2.39], $p = 0.38$

7-day model (95 % CI) 28-day model (95 % CI) runs within $\pm 15\%$ SS



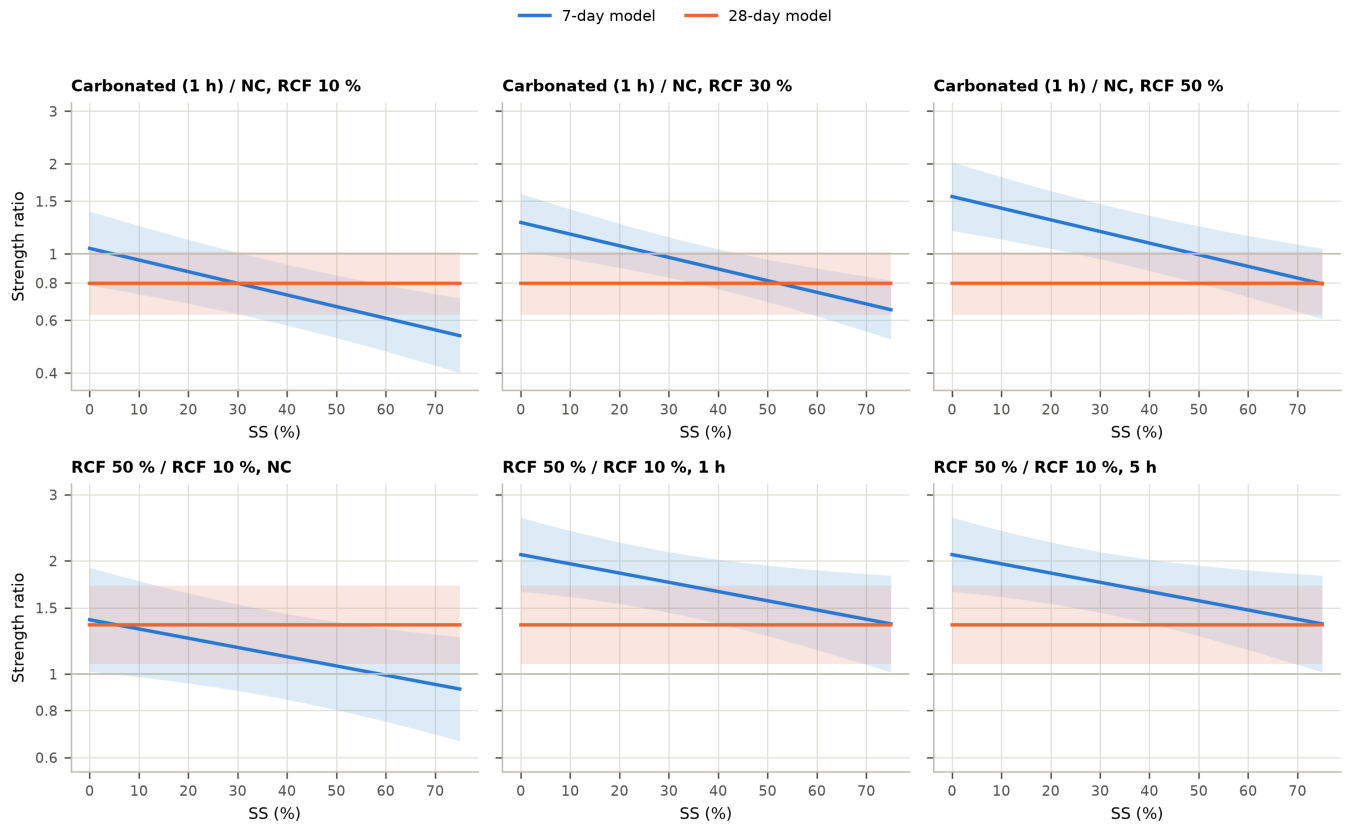
Carbonated (1 h) / NC

RCF (%)	SS (%)	7-day model: 1 h / NC	28-day model: 1 h / NC	Supplementary: 28-day data refitted with the 7-day terms
10	0	$\times 1.04$ [0.78–1.39], $p = 0.76$	$\times 0.80$ [0.63–1.01], $p = 0.064$	$\times 0.95$ [0.53–1.71], $p = 0.86$
10	25	$\times 0.83$ [0.66–1.06], $p = 0.13$	$\times 0.80$ [0.63–1.01], $p = 0.064$	$\times 0.85$ [0.52–1.40], $p = 0.51$
10	50	$\times 0.67$ [0.52–0.85], $p = 0.002$	$\times 0.80$ [0.63–1.01], $p = 0.064$	$\times 0.77$ [0.47–1.26], $p = 0.28$
10	75	$\times 0.53$ [0.40–0.71], $p < 0.001$	$\times 0.80$ [0.63–1.01], $p = 0.064$	$\times 0.69$ [0.38–1.25], $p = 0.21$
30	0	$\times 1.27$ [1.02–1.59], $p = 0.033$	$\times 0.80$ [0.63–1.01], $p = 0.064$	$\times 1.01$ [0.64–1.60], $p = 0.95$
30	25	$\times 1.02$ [0.87–1.20], $p = 0.82$	$\times 0.80$ [0.63–1.01], $p = 0.064$	$\times 0.91$ [0.65–1.27], $p = 0.57$
30	50	$\times 0.81$ [0.69–0.96], $p = 0.016$	$\times 0.80$ [0.63–1.01], $p = 0.064$	$\times 0.82$ [0.59–1.15], $p = 0.23$
30	75	$\times 0.65$ [0.52–0.82], $p < 0.001$	$\times 0.80$ [0.63–1.01], $p = 0.064$	$\times 0.74$ [0.46–1.18], $p = 0.19$
50	0	$\times 1.55$ [1.19–2.02], $p = 0.002$	$\times 0.80$ [0.63–1.01], $p = 0.064$	$\times 1.08$ [0.63–1.87], $p = 0.77$
50	25	$\times 1.24$ [1.00–1.54], $p = 0.049$	$\times 0.80$ [0.63–1.01], $p = 0.064$	$\times 0.97$ [0.62–1.52], $p = 0.90$
50	50	$\times 0.99$ [0.80–1.24], $p = 0.95$	$\times 0.80$ [0.63–1.01], $p = 0.064$	$\times 0.88$ [0.56–1.38], $p = 0.55$
50	75	$\times 0.79$ [0.61–1.04], $p = 0.091$	$\times 0.80$ [0.63–1.01], $p = 0.064$	$\times 0.79$ [0.45–1.38], $p = 0.38$

Carbonation duration

Contrast (RCF 30, SS 37.5)	7-day model	28-day model	Supplementary: 28-day data refitted with the 7-day terms
1 h / 0.5 h	$\times 0.87$ [0.71–1.05], $p = 0.14$	$\times 0.58$ [0.41–0.82], $p = 0.004$	$\times 0.79$ [0.53–1.18], $p = 0.24$
5 h / 0.5 h	$\times 1.01$ [0.84–1.22], $p = 0.93$	$\times 0.59$ [0.44–0.80], $p = 0.002$	$\times 0.73$ [0.49–1.07], $p = 0.10$
5 h / 1 h	$\times 1.16$ [1.00–1.36], $p = 0.053$	$\times 1.02$ [0.79–1.30], $p = 0.89$	$\times 0.92$ [0.67–1.26], $p = 0.58$

A/B 0.42 / 0.48: 7-day model 1 (not in model); 28-day model $\times 1.37$ [1.06–1.76], $p = 0.017$.



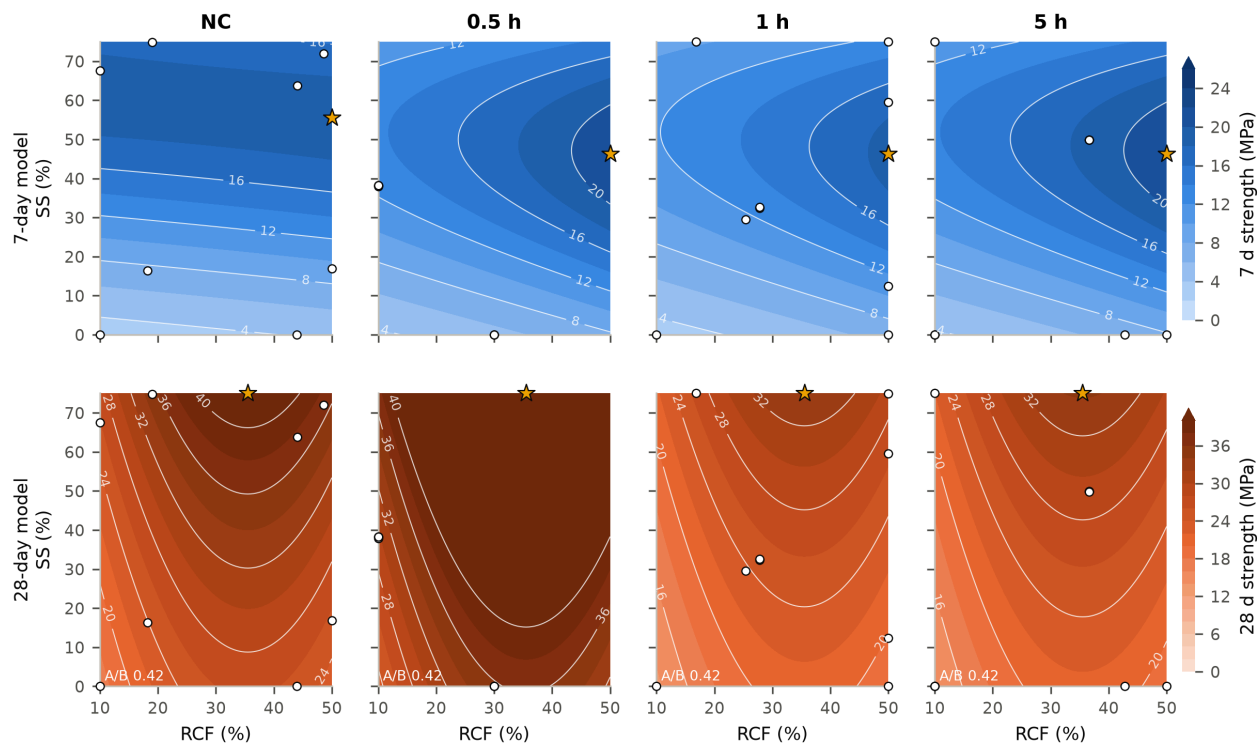
Bands: 95 % CI of the ratio. A ratio of 1 means no effect; a flat line at 1 means the term is not in that model.

Best combination per carbonation level

Carbonation	7 d: maximum at	7 d: median [95 % CI] MPa	7 d: 95 % PI	28 d: maximum at	28 d: median [95 % CI] MPa	28 d: 95 % PI
NC	RCF 50, SS 56	19.0 [15.7–23.0]	13.3–27.3	RCF 36, SS 75, A/B 0.42	42.2 [31.3–57.0]	24.2–73.9
0.5 h	RCF 50, SS 46	21.6 [17.3–27.1]	14.8–31.6	RCF 36, SS 75, A/B 0.42	58.0 [37.9–88.8] †	30.7–109.6
1 h	RCF 50, SS 46	18.8 [16.2–21.8]	13.3–26.4	RCF 36, SS 75, A/B 0.42	33.7 [26.0–43.7]	19.7–57.8
5 h	RCF 50, SS 46	21.8 [18.2–26.1]	15.3–31.1	RCF 36, SS 75, A/B 0.42	34.2 [25.0–46.9]	19.4–60.4

† Outside the region the data support (leverage above the largest design leverage): 28 d, 0.5 h. Treat these maxima as extrapolations. The 28-day maximum sits at the edge of the range of SS (no SS² term) and A/B (A/B² is positive, so A/B has a minimum inside the range).

Median predicted strength over RCF × SS from each model (own colour scale per age). ○ runs at that level, ★ maximum



Validation

